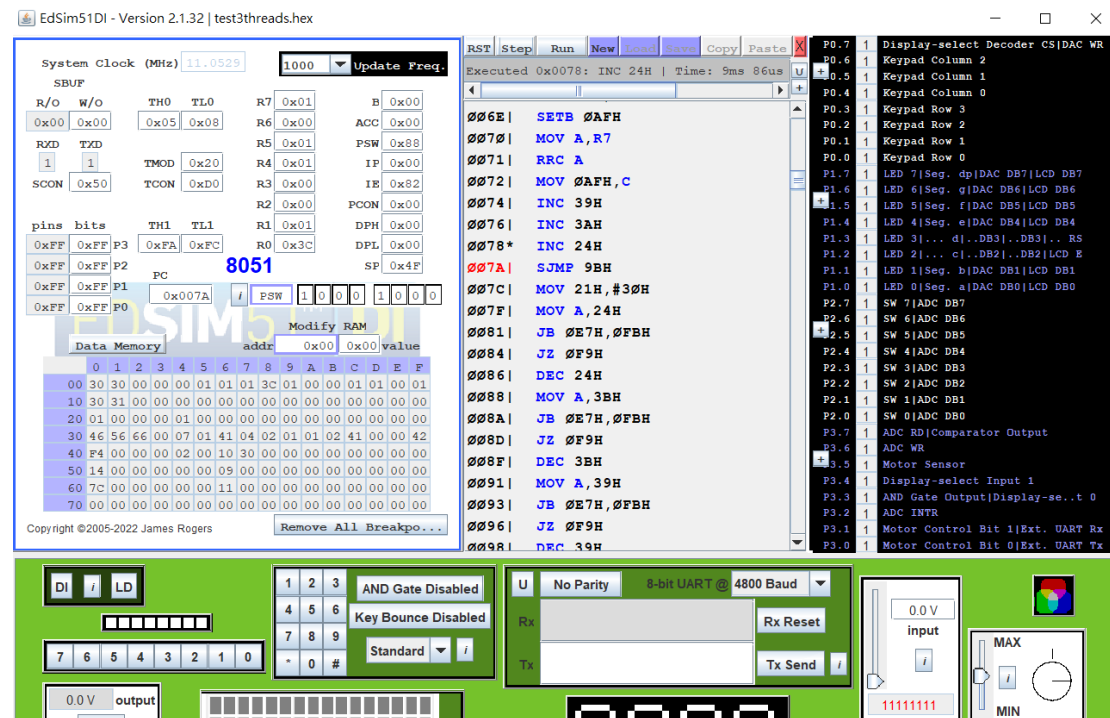
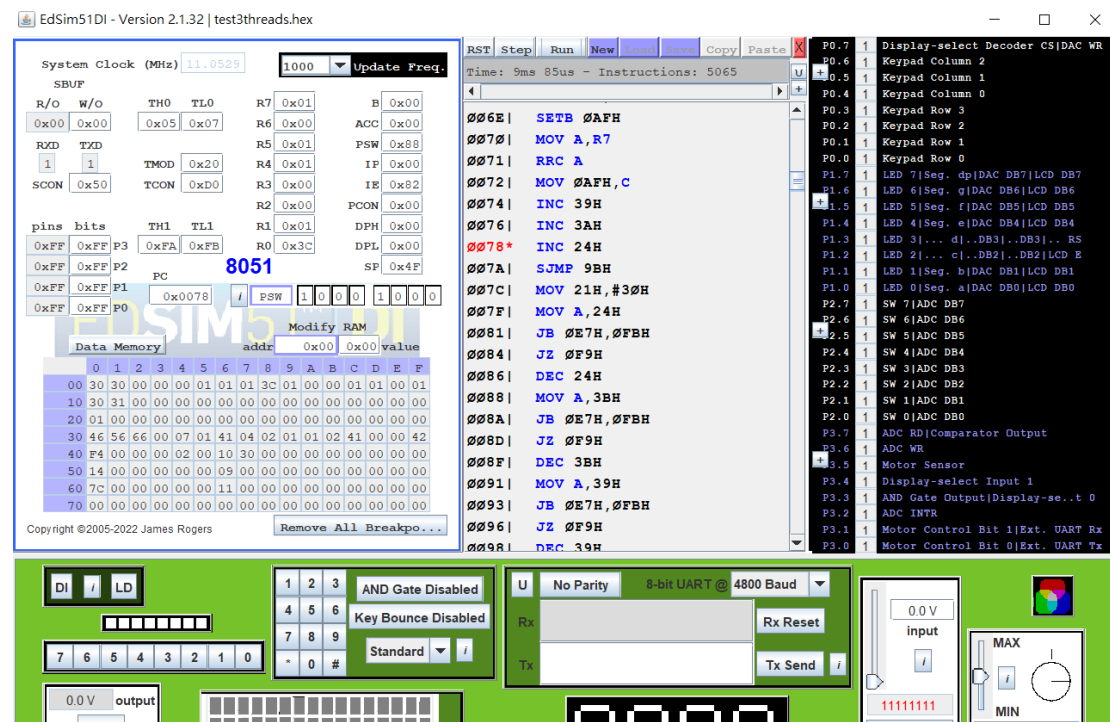


Name: 張哲語

Student ID: 109062336



Producer1 is running since the semaphore for preventing starvation "ok2" (at 0x24), (which is used to unlock Producer2) gets increased from 0 to 1.

EdSim51DI - Version 2.1.32 | test3threads.hex

System Clock (MHz) 11.0529 1000 Update Freq.

SBUF

R/O	W/O	TH0	TL0	R7	0x01	B	0x00
0x00	0x00	0x05	0x0A	R6	0x00	ACC	0x00
RxD	TxD	TMOD	0x20	R5	0x02	PSW	0x90
1	1	TCON	0xD0	R4	0x02	IP	0x00
SCON	0x50			R3	0x00	IE	0x82
				R2	0x00	PCON	0x00
pins	bits	TH1	TL1	R1	0x31	DPH	0x00
0xFF	0xFF	P3	0xFA	0xFA	0x00	DPL	0x00
0xFF	0xFF	P2					
0xFF	0xFF	P1					
0xFF	0xFF	P0					

PC 0x00E0 8051 PSW 1 0 0 1 0 0 0 0

Data Memory

0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
00	30	30	00	00	00	01	01	01	31	32	00	00	01	01	00
10	3D	31	00	00	02	02	00	01	00	00	00	00	00	00	00
20	02	31	00	00	00	00	00	00	00	00	00	00	00	00	00
30	46	56	66	00	07	02	41	04	02	01	02	01	41	30	00
40	F4	00	00	00	02	00	10	30	00	00	00	00	00	00	00
50	1C	00	00	00	00	00	88	31	01	00	00	01	01	00	00
60	7C	00	00	00	00	00	11	00	00	00	00	00	00	00	00
70	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00

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Remove All Breakpo...

DI I LD

1 2 3 AND Gate Disabled

U No Parity 8-bit UART @ 4800 Baud

0.0 V

Assembly Code:

```

00CB MOV R4,20H
00CD INC R4
00CE MOV A,R4
00CF MOV R5,A
00D0 RLC A
00D1 SUBB A,0E0H
00D3 MOV R6,A
00D4 MOV 20H,R5
00D6 SETB 0AFH
00D8 MOV A,R7
00D9 RRC A
00DA MOV 0AFH,C
00DC INC 39H
00DE INC 3AH
00E0* INC 23H
00E2 SJMP 9BH
00E4 ORL 89H,#20H
00E7 MOV 8DH,#0FAH
00EA MOV 98H,#50H
00ED SETB 8EH
00EF MOV 22H,#00H

```

IO Pins:

- P0.7 Display-select Decoder CS|DAC WR
- P0.6 Keypad Column 2
- P0.5 Keypad Column 1
- P0.4 Keypad Column 0
- P0.3 Keypad Row 3
- P0.2 Keypad Row 2
- P0.1 Keypad Row 1
- P0.0 Keypad Row 0
- P1.7 LED 7|Seg. dp|DAC DB7|LCD DB7
- P1.6 LED 6|Seg. g|DAC DB6|LCD DB6
- P1.5 LED 5|Seg. f|DAC DB5|LCD DB5
- P1.4 LED 4|Seg. e|DAC DB4|LCD DB4
- P1.3 LED 3|... d|...DB3|...DB3|...RS
- P1.2 LED 2|... c|...DB2|...DB2|LCD E
- P1.1 LED 1|Seg. b|DAC DB1|LCD DB1
- P1.0 LED 0|Seg. a|DAC DB0|LCD DB0
- P2.7 SW 7|ADC DB7
- P2.6 SW 6|ADC DB6
- P2.5 SW 5|ADC DB5
- P2.4 SW 4|ADC DB4
- P2.3 SW 3|ADC DB3
- P2.2 SW 2|ADC DB2
- P2.1 SW 1|ADC DB1
- P2.0 SW 0|ADC DB0
- P3.7 ADC RD|Comparator Output
- P3.6 ADC WR
- P3.5 Motor Sensor
- P3.4 Display-select Input 1
- P3.3 AND Gate Output|Display-se..t 0
- P3.2 ADC INTR
- P3.1 Motor Control Bit 1|Ext. UART Rx
- P3.0 Motor Control Bit 0|Ext. UART Tx

EdSim51DI - Version 2.1.32 | test3threads.hex

System Clock (MHz) 11.0529 1000 Update Freq.

SBUF

R/O	W/O	TH0	TL0	R7	0x01	B	0x00
0x00	0x00	0x05	0x0B	R6	0x00	ACC	0x00
RxD	TxD	TMOD	0x20	R5	0x02	PSW	0x90
1	1	TCON	0xD0	R4	0x02	IP	0x00
SCON	0x50			R3	0x00	IE	0x82
				R2	0x00	PCON	0x00
pins	bits	TH1	TL1	R1	0x31	DPH	0x00
0xFF	0xFF	P3	0xFA	0xFA	0x00	DPL	0x00
0xFF	0xFF	P2					
0xFF	0xFF	P1					
0xFF	0xFF	P0					

PC 0x00E2 8051 PSW 1 0 0 1 0 0 0 0

Data Memory

0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
00	30	30	00	00	00	01	01	01	31	32	00	00	01	01	00
10	3D	31	00	00	02	02	00	01	00	00	00	00	00	00	00
20	02	31	00	01	00	00	00	00	00	00	00	00	00	00	00
30	46	56	66	00	07	02	41	04	02	01	02	01	41	30	00
40	F4	00	00	00	02	00	10	30	00	00	00	00	00	00	00
50	1C	00	00	00	00	00	88	31	01	00	00	01	01	00	00
60	7C	00	00	00	00	00	11	00	00	00	00	00	00	00	00
70	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00

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Remove All Breakpo...

DI I LD

1 2 3 AND Gate Disabled

U No Parity 8-bit UART @ 4800 Baud

0.0 V

Assembly Code:

```

00CB MOV R4,20H
00CD INC R4
00CE MOV A,R4
00CF MOV R5,A
00D0 RLC A
00D1 SUBB A,0E0H
00D3 MOV R6,A
00D4 MOV 20H,R5
00D6 SETB 0AFH
00D8 MOV A,R7
00D9 RRC A
00DA MOV 0AFH,C
00DC INC 39H
00DE INC 3AH
00E0* INC 23H
00E2 SJMP 9BH
00E4 ORL 89H,#20H
00E7 MOV 8DH,#0FAH
00EA MOV 98H,#50H
00ED SETB 8EH
00EF MOV 22H,#00H

```

IO Pins:

- P0.7 Display-select Decoder CS|DAC WR
- P0.6 Keypad Column 2
- P0.5 Keypad Column 1
- P0.4 Keypad Column 0
- P0.3 Keypad Row 3
- P0.2 Keypad Row 2
- P0.1 Keypad Row 1
- P0.0 Keypad Row 0
- P1.7 LED 7|Seg. dp|DAC DB7|LCD DB7
- P1.6 LED 6|Seg. g|DAC DB6|LCD DB6
- P1.5 LED 5|Seg. f|DAC DB5|LCD DB5
- P1.4 LED 4|Seg. e|DAC DB4|LCD DB4
- P1.3 LED 3|... d|...DB3|...DB3|...RS
- P1.2 LED 2|... c|...DB2|...DB2|LCD E
- P1.1 LED 1|Seg. b|DAC DB1|LCD DB1
- P1.0 LED 0|Seg. a|DAC DB0|LCD DB0
- P2.7 SW 7|ADC DB7
- P2.6 SW 6|ADC DB6
- P2.5 SW 5|ADC DB5
- P2.4 SW 4|ADC DB4
- P2.3 SW 3|ADC DB3
- P2.2 SW 2|ADC DB2
- P2.1 SW 1|ADC DB1
- P2.0 SW 0|ADC DB0
- P3.7 ADC RD|Comparator Output
- P3.6 ADC WR
- P3.5 Motor Sensor
- P3.4 Display-select Input 1
- P3.3 AND Gate Output|Display-se..t 0
- P3.2 ADC INTR
- P3.1 Motor Control Bit 1|Ext. UART Rx
- P3.0 Motor Control Bit 0|Ext. UART Tx

Producer2 is running since the semaphore for preventing starvation “ok1”(at 0x23), which is used to unlock Producer1) gets increased from 0 to 1.

EdSim51DI - Version 2.1.32 | test3threads.hex

The screenshot displays the EdSim51DI simulation environment. The top panel shows the system clock at 11.0529 MHz and a 1000 Hz update frequency. The register window on the left lists various registers including R0-R7, B, ACC, PSW, IP, IE, PCON, DPH, DPL, SP, and PC. The PC register is highlighted at 0x0134. The memory window shows a data memory dump with addresses 0x00 to 0x7F. The instruction stream on the right shows the execution of instructions from 0x121 to 0x147. The bottom panel features a hardware interface with a keypad, a 4-digit display showing '8888', an ADC input set to 0.0V, and a motor control section. The status bar at the bottom indicates the execution time is 26ms 873us and the instruction count is 14924.

System Clock (MHz): 11.0529 | 1000 Update Freq.

SBUF: R/O W/O TH0 TL0 R7 0x01 B 0x00
R/O W/O TH0 TL0 R6 0x00 ACC 0x00
R/O W/O TH0 TL0 R5 0x01 PSW 0x90
R/O W/O TH0 TL0 R4 0x01 IP 0x00
R/O W/O TH0 TL0 R3 0x00 IE 0x82
R/O W/O TH0 TL0 R2 0x00 PCON 0x00
R/O W/O TH0 TL0 R1 0x3C DPH 0x00
R/O W/O TH0 TL0 R0 0x32 DPL 0x02
R/O W/O TH0 TL0 SP 0x3F

PC: 0x0134

Modify RAM: addr 0x00 value 0x00

Data Memory:

0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
00	30	30	00	00	00	01	01	01	31	32	00	00	01	01	00
10	32	3C	00	00	01	01	00	01	00	00	00	00	00	00	00
20	02	31	01	01	00	00	00	00	00	00	00	00	00	00	00
30	46	56	66	00	07	00	41	04	02	01	01	01	41	30	00
40	F4	00	00	00	02	00	10	30	00	00	00	00	00	00	00
50	1C	00	00	00	00	00	88	31	01	00	00	01	01	00	01
60	7F	00	00	00	00	00	90	32	31	00	00	02	02	00	01
70	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00

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Remove All Breakpo...

0121| MOV R4,22H
0123| INC R4
0124| MOV A,R4
0125| MOV R5,A
0126| RLC A
0127| SUBB A,0E0H
0129| MOV R6,A
012A| MOV 22H,R5
012C| SETB 0AFH
012E| MOV A,R7
012F| RRC A
0130| MOV 0AFH,C
0132| INC 39H
0134*| INC 3BH
0136| JBC 99H,0BBH
0139| SJMP 0FBH
013B| MOV 3AH,#00H
013E| MOV 39H,#01H
0141| MOV 3BH,#03H
0144| MOV 23H,#01H
0147| MOV 24H,#00H

Display-select Decoder CS|DAC WR
00.6| Keypad Column 2
00.5| Keypad Column 1
00.4| Keypad Column 0
00.3| Keypad Row 3
00.2| Keypad Row 2
00.1| Keypad Row 1
00.0| Keypad Row 0
P1.7| LED 7|Seg. dp|DAC DB7|LCD DB7
P1.6| LED 6|Seg. g|DAC DB6|LCD DB6
P1.5| LED 5|Seg. f|DAC DB5|LCD DB5
P1.4| LED 4|Seg. e|DAC DB4|LCD DB4
P1.3| LED 3|... d|..DB3|..DB3|.. RS
P1.2| LED 2|... c|..DB2|..DB2|LCD E
P1.1| LED 1|Seg. b|DAC DB1|LCD DB1
P1.0| LED 0|Seg. a|DAC DB0|LCD DB0
P2.7| SW 7|ADC DB7
P2.6| SW 6|ADC DB6
P2.5| SW 5|ADC DB5
P2.4| SW 4|ADC DB4
P2.3| SW 3|ADC DB3
P2.2| SW 2|ADC DB2
P2.1| SW 1|ADC DB1
P2.0| SW 0|ADC DB0
P3.7| ADC RD|Comparator Output
P3.6| ADC WR
P3.5| Motor Sensor
P3.4| Display-select Input 1
P3.3| AND Gate Output|Display-se.t 0
P3.2| ADC INTR
P3.1| Motor Control Bit 1|Ext. UART Rx
P3.0| Motor Control Bit 0|Ext. UART Tx

DI / LD
AND Gate Disabled
Key Bounce Disabled
Standard
0.0V output
Scope
DAC
BF 0 AC 0x00 IR 0x00 DR 0x00
U No Parity 8-bit UART @ 4800 Baud
Rx Reset
Tx Send
0.0V input
11111111
ADC
MAX
MIN
Motor Enabled

EdSim51DI - Version 2.1.32 | test3threads.hex

System Clock (MHz): 11.0529 | 1000 Update Freq.

SBUF: R/O W/O TH0 TL0 R7 0x01 B 0x00
R/O W/O TH0 TL0 R6 0x00 ACC 0x00
R/O W/O TH0 TL0 R5 0x01 PSW 0x90
R/O W/O TH0 TL0 R4 0x01 IP 0x00
R/O W/O TH0 TL0 R3 0x00 IE 0x82
R/O W/O TH0 TL0 R2 0x00 PCON 0x00
R/O W/O TH0 TL0 R1 0x3C DPH 0x00
R/O W/O TH0 TL0 R0 0x32 DPL 0x02
R/O W/O TH0 TL0 SP 0x3F

PC: 0x0134

Modify RAM: addr 0x00 value 0x00

Data Memory:

0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
00	30	30	00	00	00	01	01	01	31	32	00	00	01	01	00
10	32	3C	00	00	01	01	00	01	00	00	00	00	00	00	00
20	02	31	01	01	00	00	00	00	00	00	00	00	00	00	00
30	46	56	66	00	07	00	41	04	02	01	01	02	41	30	00
40	F4	00	00	00	02	00	10	30	00	00	00	00	00	00	00
50	1C	00	00	00	00	00	88	31	01	00	00	01	01	00	01
60	7F	00	00	00	00	00	90	32	31	00	00	02	02	00	01
70	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00

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Remove All Breakpo...

0121| MOV R4,22H
0123| INC R4
0124| MOV A,R4
0125| MOV R5,A
0126| RLC A
0127| SUBB A,0E0H
0129| MOV R6,A
012A| MOV 22H,R5
012C| SETB 0AFH
012E| MOV A,R7
012F| RRC A
0130| MOV 0AFH,C
0132| INC 39H
0134*| INC 3BH
0136| JBC 99H,0BBH
0139| SJMP 0FBH
013B| MOV 3AH,#00H
013E| MOV 39H,#01H
0141| MOV 3BH,#03H
0144| MOV 23H,#01H
0147| MOV 24H,#00H

Display-select Decoder CS|DAC WR
00.6| Keypad Column 2
00.5| Keypad Column 1
00.4| Keypad Column 0
00.3| Keypad Row 3
00.2| Keypad Row 2
00.1| Keypad Row 1
00.0| Keypad Row 0
P1.7| LED 7|Seg. dp|DAC DB7|LCD DB7
P1.6| LED 6|Seg. g|DAC DB6|LCD DB6
P1.5| LED 5|Seg. f|DAC DB5|LCD DB5
P1.4| LED 4|Seg. e|DAC DB4|LCD DB4
P1.3| LED 3|... d|..DB3|..DB3|.. RS
P1.2| LED 2|... c|..DB2|..DB2|LCD E
P1.1| LED 1|Seg. b|DAC DB1|LCD DB1
P1.0| LED 0|Seg. a|DAC DB0|LCD DB0
P2.7| SW 7|ADC DB7
P2.6| SW 6|ADC DB6
P2.5| SW 5|ADC DB5
P2.4| SW 4|ADC DB4
P2.3| SW 3|ADC DB3
P2.2| SW 2|ADC DB2
P2.1| SW 1|ADC DB1
P2.0| SW 0|ADC DB0
P3.7| ADC RD|Comparator Output
P3.6| ADC WR
P3.5| Motor Sensor
P3.4| Display-select Input 1
P3.3| AND Gate Output|Display-se.t 0
P3.2| ADC INTR
P3.1| Motor Control Bit 1|Ext. UART Rx
P3.0| Motor Control Bit 0|Ext. UART Tx

DI / LD
AND Gate Disabled
Key Bounce Disabled
Standard
0.0V output
Scope
DAC
BF 0 AC 0x00 IR 0x00 DR 0x00
U No Parity 8-bit UART @ 4800 Baud
Rx Reset
Tx Send
0.0V input
11111111
ADC
MAX
MIN
Motor Enabled

Consumer is running since the semaphore “empty”(at 0x3B) gets increased from 1 to 2.

U	No Parity	8-bit UART @ 4800 Baud	▼
Rx	A0B1C2D3E4F5G6H7I8J9K0L1M2N3O		
Tx			
	Rx Reset		
	Tx Send		i

Fair version: Producer1 and Producer2 both have an additional mutex ok1 and ok2, which ok1 is initialized to 1 and ok2 is initialized to 0. So after Producer1 has produced a character, it can't produce the following one until Producer2 increases ok1, so this ensures that two producers will produce characters and numbers one by one, without anyone having starvation.